Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/07/2025
PI Name:	Sunderland, Peter Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels -- Sunderland (80NSSC20M0119)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	301-405-3095
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Zip Code:	20742	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/01/2015	**End Date:**	02/08/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:	ISS		
Flight Assignment:	NOTE: Grant 80NSSC20M0119 has been extended to 02/08/2026 under new PI, Peter Sunderland (Ed., 4/11/25).		
Key Personnel Changes/Previous PI:	NOTE: Co-Investigator Peter B. Sunderland, Ph.D., is the new Principal Investigator (PI) for this investigation. The former PI, Dr. James Quintiere, passed away in December 2024. Co-Investigator John deRis has retired (Ed., 4/11/25).		
COI Name (Institution):			
Grant/Contract No.:	80NSSC20M0119		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Experimental Investigation of Emulated Burning Rate at Various Gravity Levels," (Grant 80NSSC20M0119) under new Principal Investigator (Ed., 4/11/25). A novel experimental apparatus, the Burning Rate Emulator (BRE), was developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of virtual solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it enables the emulation of steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements were conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard		

Task Description:	the International Space Station. The Burning Rate Emulator (BRE) simulated the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE controlled the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This allowed it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. Extinguishment was considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) were considered. The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows. 1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21% oxygen: 10.2 psia, 26.5%; and 8.2 psia, 34% oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels. 2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired). 3. Consider the conditions of Objectives 1 – 2 for extinction limits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	We continued working on the data analysis and publication of all the Burning Rate Emulator (BRE) data. In the coming year, we plan to continue our analysis of the International Space Station (ISS) flight tests. This abstract was submitted: P. Dehghani, P.B. Sunderland, J.G. Quintiere, Measurements and Modeling of Microgravity Pool Fire Temperature, U.S. National Combustion Meeting, Boston, Massachusetts, abstract, 2025. This oral presentation was made: P. Dehghani, Measurements and Modeling of Microgravity Pool Fire Temperature, U.S. National Combustion Meeting, Boston, Massachusetts, March 19, 2025. This poster was presented: P. Dehghani, P.B. Sunderland, J.G. Quintiere, Microgravity Pool Fire Temperature Measurements and Modeling, Poster, ISFEH, Rome, 2025. Note: See Cumulative Bibliography (Ed., 10/8/25).
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Papers from Meeting Proceedings	Dehghani P, Sunderland PB, Quintiere JG. "Microgravity pool fire temperature measurements and modeling." 11th International Seminar on Fire and Explosion Hazards (ISFEH), Rome, Italy, June 15-20, 2025. Poster. Abstracts. 11th International Seminar on Fire and Explosion Hazards (ISFEH), Rome, Italy, June 15-20, 2025. , Jun-2025